

Behavior-Aware Consistent Distillation for Cold-Start Recommendation

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Abstract—Cold-start recommendation for users and items with no prior interactions remains a fundamental challenge, as content-driven cold-start entities differ substantially from behavior-rich warm entities. Generative methods attempt to address this by inferring embeddings from content features, while dropout-based techniques improve robustness by randomly masking behavioral signals during training. However, these approaches primarily focus on limited cold-start scenarios and do not fully capture the differences between cold-start and warm recommendations. As a result, generative models may produce unbalanced predictions and dropout strategies can reduce recommendation quality for warm entities. To address these limitations, we propose Behavior-Aware Consistent Distillation (BACD), a unified framework for cold-start recommendation for both users and items. BACD employs a behavior-informed teacher-student paradigm together with a behavior-aware teacher-quality weighting mechanism to ensure reliable supervision. In addition, it introduces three consistency-oriented distillation objectives, including Rating Distribution Consistency, Ranking Consistency, and Representation Distribution Consistency, which collectively reduce the gap between cold-start and warm entities. Extensive experiments on four benchmark datasets show that BACD consistently improves recommendation performance in both cold-start and overall settings and achieves

substantial gains over state-of-the-art baselines across multiple backbone models.

Index Terms—Cold-start recommendation, consistent distillation, content features.

I. INTRODUCTION

EMBEDDING-BASED collaborative filtering (CF) techniques have established new benchmarks in recommendation research by efficiently capturing complex interaction patterns between users and items through learned representations derived from past behaviors [1], [2], [3], [4], [5]. Traditional CF algorithms primarily aim to generate embeddings that enhance predictive performance using dot product operations or multilayer perceptrons (MLPs) applied to historically interacting user-item pairs [1], [2]. In contrast, graph-based CF models leverage neighborhood relationships to enrich user and item representations, which further improves recommendation accuracy [3], [4], [6]. These embedding-based approaches generally perform well when both users and items have sufficient interaction histories. However, the rapid emergence of cold-start scenarios, such as newly joined users or recently published content including live streams [7], [8] and short-form videos [9], [10], poses new challenges. The scarcity of prior behavioral data in these cases limits recommendation accuracy, making it difficult to produce reliable predictions and potentially reducing both user engagement and system profitability.

Addressing these challenges, recent recommendation frameworks aim to construct dependable embeddings for users and items with little or no interaction data while fostering mutual reinforcement between behavior-rich and behavior-sparse entities [11], [12], [13]. Generative approaches, in particular, synthesize meaningful user-item interaction representations for cold-start cases, thereby enhancing overall system-level recommendation effectiveness. For example, DeepMusic [14] reduces the mean squared error (MSE) between synthetic embeddings and empirically learned embeddings of active (warm) items. Similarly, meta-learning strategies [12], [13] adopt few-shot paradigms to transfer content-based features of cold-start entities into embedding spaces that rapidly align with those of warm counterparts. In contrast, embedding dropout-based methods improve model resilience by stochastically removing behavioral signals during training. Notable examples include DropoutNet [15] and Heater [11], which apply random dropout to embedding vectors, as well as CLCRec [16], which employs

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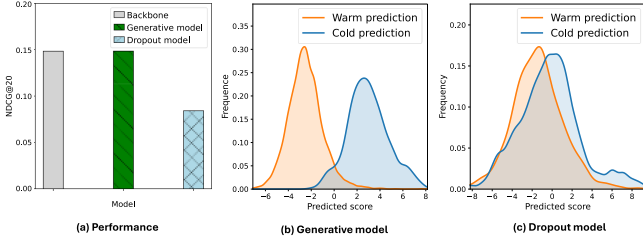


Fig. 1. Comparison of generative and dropout approaches on CiteULike: (a) warm recommendation effectiveness, (b) distribution of cold vs. warm item predictions using DeepMusic (generative), and (c) distribution of cold vs. warm item predictions using DropoutNet (dropout).

contrastive loss to mitigate the distributional gap in predicted scores between cold and warm items. Despite their strengths in cold-start representation learning, generative frameworks often struggle to maintain consistency between cold-start and warm entities within a unified ranking space. As illustrated in Fig. 1(a), these models typically rely on separate processing pathways for different entity types, resulting in output distributions that reflect divergent training conditions. This separation can lead to score mismatches, over-exposing low-quality cold-start content or under-representing emerging users due to conservative scoring. Conversely, dropout-based strategies promote more coherent treatment across entity types, as shown in Fig. 1(c), producing more balanced rating distributions across cold and warm cases. However, this benefit may come at the cost of reduced precision for behavior-rich entities, potentially diminishing the continued visibility of high-quality or popular content.

The root of the cold-start challenge lies in the intrinsic discrepancy between recommendation strategies based on historical user-item interactions (used for warm users and items) and those relying solely on content-based features (necessary for cold-start users or items). Although recent generative and dropout-based techniques have improved cold-start performance to some extent, they still struggle to bridge the behavioral gap between content-driven and interaction-driven representations. A promising approach is recommendation-oriented knowledge distillation, where behavioral knowledge from warm users and items is transferred to guide the learning of cold-start entities. Teacher models operate on behavior-rich entities, while student models learn to approximate similar recommendation behaviors from content features alone. Existing distillation methods, such as Rank Distillation [17], [18] and Collaborative Distillation [19], have shown effectiveness in transferring knowledge. However, applying them in cold-start recommendation presents several challenges:

- 1) *Unified Handling of Users and Items*: Cold-start scenarios involve both new users and new items. Many methods focus on item-only or user-only settings. Effective transfer requires a unified framework that can simultaneously handle cold-start entities on both sides of the interaction.
- 2) *Variable Teacher Reliability*: Warm entities may vary in the quality of their behavioral patterns. Distilling knowledge from unreliable teachers can introduce noise and degrade performance, making it essential to evaluate and

weight teacher contributions based on their behavioral consistency.

- 3) *Behavioral Discrepancy Modeling*: Cold-start entities rely on content-based embeddings, whereas warm entities rely on behavioral signals, resulting in inherent differences in predictions. Capturing and mitigating this discrepancy is critical for reliable knowledge transfer.

To address these issues, we propose **Behavior-Aware Consistent Distillation (BACD)**, a unified framework that improves cold-start recommendation for both users and items. BACD leverages behavior-rich entities as teachers to guide content-driven student models while employing a **behavior-aware teacher-quality weighting mechanism** to selectively emphasize reliable supervision. Within this framework, three complementary distillation objectives are applied: **rating distribution loss**, **ranking consistency loss**, and **representation distribution consistency loss**, which together enhance prediction consistency and representation quality for cold-start entities.

Our main contributions are summarized as follows:

- We propose a unified distillation framework for cold-start recommendation that simultaneously addresses both new users and new items, ensuring consistent predictive and representational behavior across all types of cold-start interactions.
- A behavior-aware teacher-quality weighting mechanism is introduced to assess warm-entity reliability and mitigate the influence of unreliable teachers during distillation.
- BACD identifies key sources of discrepancy between teacher and student predictions and designs corresponding distillation objectives: Rating Distribution Consistency, Ranking Consistency, and Representation Distribution Consistency, which jointly promote robust knowledge transfer.
- Extensive experiments on four public datasets covering both cold-start items and cold-start users demonstrate that BACD consistently outperforms existing baselines and prior distillation methods under various backbone architectures.

This paper is an extended version of our conference paper [20], which significantly advances the original framework through three key dimensions: **generality**, **methodology**, and **empirical validation**. For **generality**, we propose BACD as a unified framework that simultaneously handles both cold-start user and cold-start item scenarios, thereby overcoming the item-only limitation of the prior work. This is enabled by dual transformation functions (f_U and f_I) together with a unified prediction mechanism that seamlessly integrates behavioral and content-based representations. For **methodology**, BACD introduces two major enhancements: (i) a behavior-aware teacher-quality weighting mechanism that dynamically evaluates and emphasizes reliable supervision from both user and item teachers, where reliability is assessed by interaction sufficiency and local embedding consistency, and (ii) a new representation distribution consistency objective that enforces global statistical agreement between teacher and student embeddings in terms of mean and covariance, which is particularly crucial for addressing cold-start users lacking behavioral data. These complement the

two inherited objectives, rating distribution consistency and ranking consistency, and together form a more comprehensive distillation scheme. For **empirical validation**, we conduct extensive experiments on four benchmark datasets and multiple backbone architectures, and the results demonstrate that BACD consistently outperforms state-of-the-art baselines as well as the earlier framework across diverse cold-start scenarios.

The remainder of this paper is organized as follows. Section II reviews related work; Sections III and IV formulate the problem and present BACD; Section V reports the experiments; and Section VI concludes the paper.

II. RELATED WORK

A. Embedding-Based Recommendation

Personalized recommendation systems aim to suggest tailored items to a vast user base, navigating through an extensive inventory of options [21], [22], [23]. A prevalent strategy in this domain involves the utilization of embedding vectors to represent individual users or items. In this context, Matrix Factorization (MF) models [2], [24], [25] are adept at decomposing the user-item interaction matrix into constituent user and item embedding vectors.

Graph Neural Networks (GNNs) [26], [27], [28], [29] have become central to graph-structured learning, supporting tasks such as node classification and link prediction. Since user-item interactions naturally form a graph, NGCF [3] incorporates graph convolution into collaborative filtering to learn informative user and item embeddings. To improve efficiency while preserving accuracy, LightGCN [4] removes non-linear layers and achieves state-of-the-art recommendation performance.

B. Cold-Start Recommendation

The cold-start problem in recommender systems, caused by the lack of sufficient interaction data for newly introduced users or items, has been studied through a variety of strategies [30], [31]. Early approaches mainly exploit content information to construct mappings from raw user/item attributes to collaborative representations. Representative examples include DeepMusic [14] and meta-learning based methods such as MetaEmb [12].

Generative adversarial learning has also been used to alleviate cold-start difficulty. Methods such as RAGAN [32], LARA [33], and GAR [34] improve cold-start recommendation by enhancing or synthesizing embeddings for new items. In addition, dropout-inspired techniques, including DropoutNet [15], MTPR [35], CC-CC [36], Heater [11], VELF [37], and CLCRec [16], simulate cold entities during training by masking or perturbing learned embeddings, thereby improving robustness.

Recent studies have explored LLMs for cold-start recommendation by leveraging rich semantic knowledge in textual content, e.g., for cold-item data augmentation [38], user-item interaction simulation [39], and retrieval-augmented cold-start user modeling [40]. These methods typically incorporate LLMs during data generation, training, or inference. Our work instead focuses on the standard collaborative/content-feature cold-start

setting, where BACD transfers behavioral knowledge from warm entities to cold entities through behavior-aware consistent distillation.

C. Knowledge Distillation

Knowledge distillation transfers knowledge from a stronger teacher model to a simpler student, often improving performance while maintaining efficiency [41]. It has been widely applied to vision, language modeling, and other machine learning tasks [42], [43], [44], [45], [46].

In recommender systems, distillation has been studied as ranking distillation [17], [19], [47], [48], adversarial distillation [49], [50], and privileged-feature distillation [51], [52], [53]. Although effective, most methods target standard warm recommendation and rarely consider cold-start settings, where users or items have limited or no interaction history. In this case, successful transfer depends not only on teacher supervision, but also on consistent alignment between warm and cold entities, motivating our behavior-aware consistent distillation framework.

III. PRELIMINARIES

A. Notations

In this paper, $\mathcal{U}$ and $\mathcal{I}$ denote the user and item sets, and $\mathcal{H}$ denotes the interactions available in the training graph unless otherwise specified. Users are partitioned into warm users $\mathcal{U}_w$ with observed training interactions and cold users $\mathcal{U}_c$ whose interactions are withheld from the training graph; items are similarly divided into $\mathcal{I}_w$ and $\mathcal{I}_c$.

For warm entities, behavioral embeddings are learned by recommendation backbones such as Matrix Factorization, Neural Matrix Factorization, and Graph Neural Networks. We denote the warm-user and warm-item embedding matrices by $E_{\mathcal{U}_w}$ and $E_{\mathcal{I}_w}$, and individual embeddings by e_u for $u \in \mathcal{U}_w$ and e_i for $i \in \mathcal{I}_w$.

For cold users and items, which do not have behavioral embeddings, BACD uses available content features such as user profiles and item descriptions to construct content-based representations, denoted by c_u and c_i for any $u \in \mathcal{U}$ and $i \in \mathcal{I}$. The main notations are summarized in Table I.

B. Cold-Start Recommendation

a) Problem Definition: The goal of a recommender system is to generate a top- K ranked list of items for each user, based on predicted relevance scores over the entire candidate set. In cold-start scenarios, the system must generate meaningful predictions even when the user, the item, or both, lack historical interaction data. To evaluate performance under various cold-start conditions, we define the following three tasks:

- *Overall Recommendation (Rec_{all}):*

$$Rec_{all} = \{\text{rank}_K(\{(i, \hat{y}_{ui}) \mid i \in \mathcal{I}\}) \mid u \in \mathcal{U}\}. \quad (1)$$

- *Warm Recommendation (Rec_w):*

$$Rec_w = \{\text{rank}_K(\{(i, \hat{y}_{ui}) \mid i \in \mathcal{I}_w\}) \mid u \in \mathcal{U}_w\}. \quad (2)$$

TABLE I
MAIN SYMBOLS AND DEFINITIONS USED IN THIS PAPER

Symbol	Description
$\mathcal{U}$	Set of users
$\mathcal{I}$	Set of items
$\mathcal{H}$	Set of user-item interactions in the training graph
H	Matrix of user-item interaction
$\mathcal{B}$	Set of preference triplets (u, i, j)
$\mathcal{B}_I$	Set of positive items in $\mathcal{B}$
$\mathcal{B}_J$	Set of negative items in $\mathcal{B}$
$\mathcal{U}_w$	Set of warm users (with observed training interactions)
$\mathcal{U}_c$	Set of cold users (without observed training interactions)
$\mathcal{I}_w$	Set of warm items (with observed training interactions)
$\mathcal{I}_c$	Set of cold items (without observed training interactions)
d	Dimension of user/item embeddings
$\mathbf{e}_u$	Embedding vector of user u
$\mathbf{e}_i$	Embedding vector of item i
$\mathbf{c}_u$	Content feature vector of user u
$\mathbf{c}_i$	Content feature vector of item i
$\hat{y}_{ui}$	Predicted interaction score between user u and item i

- *Cold Recommendation (Rec_c):*

- *Cold Item Recommendation (CIR):*

$$Rec_{c-i} = \{\text{rank}_K(\{(i, \hat{y}_{ui}) \mid i \in \mathcal{I}_c\}) \mid u \in \mathcal{U}_w\}. \quad (3)$$

- *Cold User Recommendation (CUR):*

$$Rec_{c-u} = \{\text{rank}_K(\{(i, \hat{y}_{ui}) \mid i \in \mathcal{I}_w\}) \mid u \in \mathcal{U}_c\}. \quad (4)$$

- *Cold User and Item Recommendation (CU&IR):*

$$Rec_{c-b} = \{\text{rank}_K(\{(i, \hat{y}_{ui}) \mid i \in \mathcal{I}_c\}) \mid u \in \mathcal{U}_c\}.$$

In all cases, the function $\text{rank}_K(\mathcal{S})$ returns the top- K items with the highest predicted scores from the score-indexed candidate set $\mathcal{S}$. These formulations allow us to independently analyze model performance on cold-start users, cold-start items, and their combinations, in addition to overall and warm-only settings.

b) Generative Models: Generative models extend conventional recommendation frameworks by generating approximate embeddings for cold-start users and items based on available content features. Typically, embeddings for warm users and items are retained, while type-specific generator functions $g_{\mathcal{U}}(\cdot)$ and $g_{\mathcal{I}}(\cdot)$ are applied to produce embeddings for cold users and cold items from content information. The prediction score for a user-item pair is computed as:

$$\hat{y}_{ui} = \begin{cases} \mathbf{e}_u^\top \cdot \mathbf{e}_i, & \text{if } u \in \mathcal{U}_w, i \in \mathcal{I}_w, \\ g_{\mathcal{U}}(\mathbf{c}_u)^\top \cdot \mathbf{e}_i, & \text{if } u \in \mathcal{U}_c, i \in \mathcal{I}_w, \\ \mathbf{e}_u^\top \cdot g_{\mathcal{I}}(\mathbf{c}_i), & \text{if } u \in \mathcal{U}_w, i \in \mathcal{I}_c, \\ g_{\mathcal{U}}(\mathbf{c}_u)^\top \cdot g_{\mathcal{I}}(\mathbf{c}_i), & \text{if } u \in \mathcal{U}_c, i \in \mathcal{I}_c. \end{cases}$$

While generative models maintain the recommendation capability for warm entities, they can produce inconsistencies in prediction distributions between warm and cold users or items, particularly affecting score ranges and ranking behaviors, as illustrated in Fig. 1(b).

c) Dropout-Based Models: Dropout-inspired approaches, such as DropoutNet [15] and Heater [11], improve robustness by simulating missing behavioral embeddings during training.

They combine behavioral embeddings and content features through mapping functions $f_{\mathcal{U}}(\cdot)$ and $f_{\mathcal{I}}(\cdot)$, yielding relevance scores:

$$\hat{y}_{ui} = \begin{cases} f_{\mathcal{U}}(\mathbf{e}_u, \mathbf{c}_u)^\top \cdot f_{\mathcal{I}}(\mathbf{e}_i, \mathbf{c}_i), & \text{if } u \in \mathcal{U}_w, i \in \mathcal{I}_w, \\ f_{\mathcal{U}}(\mathbf{0}, \mathbf{c}_u)^\top \cdot f_{\mathcal{I}}(\mathbf{e}_i, \mathbf{c}_i), & \text{if } u \in \mathcal{U}_c, i \in \mathcal{I}_w, \\ f_{\mathcal{U}}(\mathbf{e}_u, \mathbf{c}_u)^\top \cdot f_{\mathcal{I}}(\mathbf{0}, \mathbf{c}_i), & \text{if } u \in \mathcal{U}_w, i \in \mathcal{I}_c, \\ f_{\mathcal{U}}(\mathbf{0}, \mathbf{c}_u)^\top \cdot f_{\mathcal{I}}(\mathbf{0}, \mathbf{c}_i), & \text{if } u \in \mathcal{U}_c, i \in \mathcal{I}_c. \end{cases}$$

By jointly training over warm and cold combinations, dropout models promote consistency across cold-start conditions. However, mapping behavioral embeddings through transformation functions may reduce the fidelity of warm predictions compared with the backbone models (Fig. 1(a)).

C. Ranking Distillation

Ranking distillation [17] transfers ranking information from a complex teacher model to a compact student model. Its objective combines the standard recommendation loss with an additional ranking-oriented term:

$$\min_{\theta_{\text{student}}} \mathcal{L}_{\text{rec}} + \lambda \mathcal{L}_{\text{RD}},$$

where θ_{student} denotes the student parameters, $\mathcal{L}_{\text{rec}}$ is the recommendation loss, and $\mathcal{L}_{\text{RD}}$ encourages the student to approximate the teacher's ranking behavior through score alignment [17], [19].

Unlike traditional model compression, where the teacher is discarded at inference, cold-start recommendation requires teacher and student predictions to coexist: the teacher handles warm entities with behavioral embeddings, while the student handles cold entities with content features. As illustrated in Fig. 2, our framework introduces ranking consistency and other consistency-based objectives to reduce the gap between warm and cold-start scenarios.

IV. METHODOLOGY

In this section, we present the detailed design of the proposed **Behavior-Aware Consistent Distillation (BACD)** framework, which addresses both cold-start user and item recommendation by ensuring consistency between content-based student models and behavior-based teacher models. We first introduce the overall architecture and teacher-student formulation, then model behavioral discrepancies through a teacher-quality weighting mechanism, and finally describe three distillation objectives that guide the student toward consistent prediction and representation across cold-start scenarios.

A. Comprehensive Framework

1) Characterization of Teachers and Students: Within the BACD framework, as illustrated in Fig. 2, the “teacher” model predicts user-item interactions using behavioral histories from both sides. In contrast, the “student” model handles cold-start cases, where the user, the item, or both lack interaction records, and thus predictions depend solely on content-based representations.

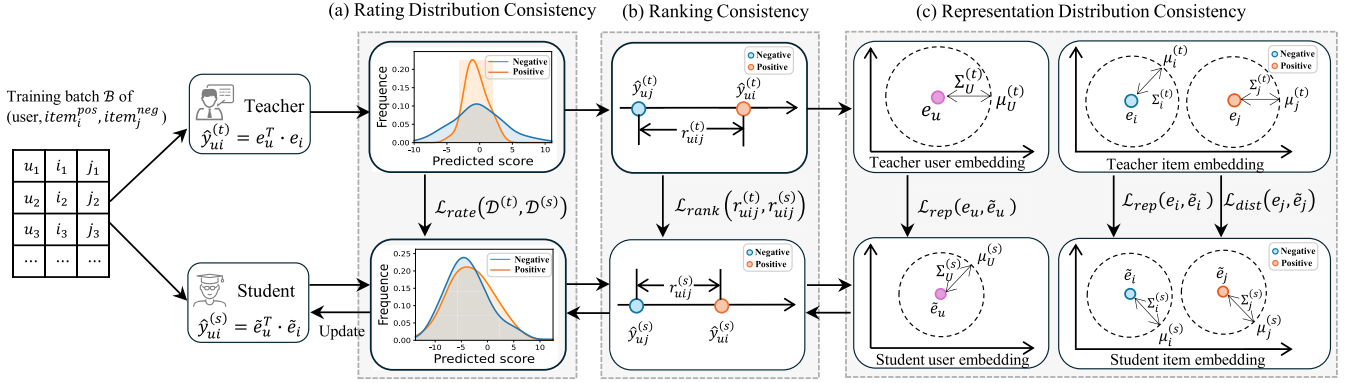


Fig. 2. An overview of the proposed BACD framework. The teacher model generates predictions using behavior-based embeddings from warm users and items, while the student model relies on content-based representations for cold-start entities. By comparing their outputs, the framework applies three consistency-driven objectives, which encompass rating distribution consistency, ranking consistency, and representation distribution consistency, to reduce behavioral discrepancies and enhance the quality of cold-start recommendations.

For a given pair (u, i) , the teacher computes the relevance score as the dot product between the user and item embeddings obtained from behavioral data:

$$\hat{y}_{ui}^{(t)} = e_u^\top e_i.$$

To support cold-start cases, the BACD framework employs two transformation functions: $f_U(\cdot)$ projects user content features c_u into the embedding space, while $f_I(\cdot)$ transforms item content features c_i . These functions are implemented as two-layer MLPs for flexibility and efficiency. The student model then computes its relevance score as:

$$\hat{y}_{ui}^{(s)} = \tilde{e}_u^\top \tilde{e}_i,$$

where

$$\tilde{e}_x = \begin{cases} e_x, & \text{if } x \in \mathcal{U}_w \cup \mathcal{I}_w, \\ f_U(c_x), & \text{if } x \in \mathcal{U}_c, \\ f_I(c_x), & \text{if } x \in \mathcal{I}_c. \end{cases}$$

Here, x can be either a user or an item. For warm entities with observed training interactions, the student reuses the behavioral embedding from the teacher; for cold users and cold items, it generates embeddings from content features using $f_U(\cdot)$ and $f_I(\cdot)$, respectively.

The final prediction for any user-item pair is defined as:

$$\hat{y}_{ui} = \begin{cases} \hat{y}_{ui}^{(t)}, & \text{if } u \in \mathcal{U}_w \text{ and } i \in \mathcal{I}_w, \\ \hat{y}_{ui}^{(s)}, & \text{otherwise.} \end{cases}$$

This unified prediction mechanism enables the framework to seamlessly handle all types of user-item interactions, including *warm user-warm item*, *warm user-cold item*, *cold user-warm item*, and *cold user-cold item* cases, by leveraging behavioral embeddings when available and reverting to content-based representations otherwise. Here, “warm” and “cold” refer to whether the user or item has historical interaction records, with “cold” indicating the absence of such interactions. As a result, BACD provides a robust and generalizable solution to dual cold-start recommendation.

2) Behavioral Discrepancy Analysis: In this subsection, we examine the intrinsic disparities between the warm-start predictions made by teachers and the cold-start predictions made

by students, discrepancies that can challenge the cohesion and reliability of the recommendation system. Without proper treatment, an imbalance in these predictions may lead to suboptimal recommendations of cold-start items or hinder the exposure of desirable ones. The cold-start process, which relies on content-based embeddings instead of behavior-driven ones, introduces three core challenges:

- i) *Discrepancy in Rating Distributions:* As shown in Fig. 2(a), teacher models yield sharp and well-separated score distributions based on rich behavioral signals. In contrast, student models, dependent on content, often produce flatter and less discriminative scores. This motivates the *rating distribution consistency loss* $\mathcal{L}_{\text{rate}}$, which encourages the student to mimic the teacher’s output distribution.
- ii) *Inconsistent Ranking Orders:* Fig. 2(b) shows that student models may rank items in a different order than teacher models, especially when identifying relevant versus irrelevant items. Such inconsistencies degrade recommendation quality and are addressed by the *ranking consistency loss* $\mathcal{L}_{\text{rank}}$, which ensures consistency in relative ranking orders between teacher and student.
- iii) *Distributional Discrepancy in Representation Spaces:* As visualized in Fig. 2(c), the global statistical structure of the student’s representation space may deviate from that of the teacher. Inconsistency in first-order (mean) and second-order (covariance) statistics can hinder generalization. We propose the *representation distribution consistency loss* $\mathcal{L}_{\text{rep}}$ to minimize this representation gap.

To jointly address these issues and enhance the student’s capability under teacher guidance, we formulate the overall training objective as:

$$\min_{\theta_f} (\mathcal{L}_{\text{basic}} + \lambda_1 \mathcal{L}_{\text{rate}} + \lambda_2 \mathcal{L}_{\text{rank}} + \lambda_3 \mathcal{L}_{\text{rep}}),$$

where θ_f denotes the parameters of the transformation functions $f_U(\cdot)$ and $f_I(\cdot)$. The base loss $\mathcal{L}_{\text{basic}}$ ensures the student acquires essential recommendation abilities, typically optimized using Bayesian Personalized Ranking (BPR). The additional three terms correspond to the proposed distillation objectives, each

facilitating a specific form of teacher-student consistency. The hyperparameters λ_1 through λ_3 balance the influence of each loss component.

3) *Construction of Training Batches*: To optimize the BACD framework, we adopt a batch-wise loss computation scheme, as illustrated in Fig. 2. Each batch is formed as a set of triplets, including positive user-item pairs from $\mathcal{H}$ and negative pairs sampled from the warm items. Formally, the batch set is defined as

$$\mathcal{B} = \{(u, i, j) \mid (u, i) \in \mathcal{H}, (u, j) \notin \mathcal{H}, i \in \mathcal{I}_w, j \in \mathcal{I}_w\},$$

where $\mathcal{H}$ denotes the observed interaction set. The following subsections describe the three distillation losses in BACD, all of which are computed over the batch set $\mathcal{B}$ to address the cold-start challenges. For clarity, we denote the corresponding user and item sets as:

$$\mathcal{B}_U = \{u \mid (u, i, j) \in \mathcal{B}\},$$

$$\mathcal{B}_I = \{i \mid (u, i, j) \in \mathcal{B}\}, \quad \mathcal{B}_J = \{j \mid (u, i, j) \in \mathcal{B}\}.$$

This batch construction mechanism plays a crucial role in evaluating the model's ability to discern user preferences and capture rating distribution patterns. By incorporating both positive and negative pairs within each batch, the model is trained to differentiate between relevant and irrelevant items, which enhances recommendation accuracy, particularly in cold-start scenarios.

B. Behavior-Aware Teacher-Quality Weighting

In the **BACD** framework, we enhance distillation by assigning teacher signals different importance according to their reliability. Unlike conventional approaches that treat all teachers as equally trustworthy, BACD recognizes that user and item behavioral embeddings may vary in quality because they are learned from historical interactions. Embeddings supported by sparse interactions or inconsistent local structures can introduce noisy supervision and degrade distillation, especially in cold-start scenarios.

To address this issue, BACD introduces a behavior-aware teacher-quality weighting mechanism for both users and items. It jointly considers interaction richness and local embedding consistency within behavior-defined neighborhoods, thereby calibrating each teacher signal in a principled manner.

Teacher-Quality Weighting for Items and Users: In **BACD**, we design a unified weighting scheme to quantify the reliability of embeddings as distillation teachers, applicable to both items and users. The reliability of a teacher embedding depends on two factors: (i) the sufficiency of its interaction history and (ii) the structural consistency of the embedding with its behavior-defined neighborhood. Intuitively, embeddings of entities with richer interaction histories are more informative, while embeddings that are locally consistent with their neighbors in the interaction graph are more trustworthy.

Interaction-based weight: For an entity x (item i or user u), we define an interaction-based weight to capture reliability from historical activity:

$$w_x^{\text{freq}} = \frac{2}{1 + \exp\left(-\alpha \cdot \frac{N_x}{\bar{N}_x}\right)} - 1,$$

where N_x is the number of interactions associated with x (number of users for item i , number of items for user u), $\bar{N}_x$ is the mean over all corresponding entities, and $\alpha > 0$ controls the growth rate.

Neighborhood consistency weight: Let $H \in \mathbb{R}^{|\mathcal{U}| \times |\mathcal{I}|}$ denote the user-item interaction matrix, where $H_{ui} = 1$ if user u has interacted with item i and 0 otherwise. Define the entity-entity similarity induced by co-occurrence as follows: for items, $\mathbf{S}^I = H^\top H \in \mathbb{R}^{|\mathcal{I}| \times |\mathcal{I}|}$, where S_{ij}^I counts the number of users who interacted with both items i and j ; for users, $\mathbf{S}^U = H H^\top \in \mathbb{R}^{|\mathcal{U}| \times |\mathcal{U}|}$, where S_{uv}^U counts the number of items both users u and v have interacted with. In practice, we do not explicitly materialize the full matrices $\mathbf{S}^I$ and $\mathbf{S}^U$. Instead, for each warm item or user, we construct a sparse top- K_{nbr} neighbor list according to the corresponding co-occurrence scores, where K_{nbr} denotes the retained neighborhood size, and compute the neighborhood consistency weight only on these retained neighbors. For each entity x (item or user), let $\mathcal{N}_x^{(1)}$ denote its top- K_{nbr} most similar 1-hop neighbors. For an item entity $x = i$ and its item neighbor $y = j$, we set $s_{xy} = S_{ij}^I$; for a user entity $x = u$ and its user neighbor $y = v$, we set $s_{xy} = S_{uv}^U$. We define the normalization term as $Z_x = \sum_{y \in \mathcal{N}_x^{(1)}} s_{xy}$. The neighborhood deviation and reliability of entity x are then defined as

$$\delta_x^{\text{neigh}} = \frac{1}{Z_x} \sum_{y \in \mathcal{N}_x^{(1)}} s_{xy} \|\mathbf{e}_x - \mathbf{e}_y\|_2^2,$$

$$w_x^{\text{neigh}} = \exp(-\beta \delta_x^{\text{neigh}}),$$

where $\beta > 0$ controls the decay rate. Here, δ_x^{neigh} measures the weighted average deviation of the embedding of x from its most similar neighbors, with weights proportional to the number of shared interactions. Smaller δ_x^{neigh} indicates stronger local consistency, meaning that the embedding of x is smooth and coherent with related entities. The exponential transformation converts this deviation into a reliability weight $w_x^{\text{neigh}} \in (0, 1]$, so that embeddings with stronger local agreement receive higher weights in distillation.

Final teacher-quality weight: The overall reliability of a teacher embedding is obtained by combining its interaction-based weight w_x^{freq} , reflecting the sufficiency of historical interactions, with the neighborhood consistency weight w_x^{neigh} , capturing local embedding smoothness:

$$w_x = w_x^{\text{freq}} \cdot w_x^{\text{neigh}}.$$

This formulation provides a principled and interpretable measure of teacher quality, balancing the quantity of interaction information and the quality of embedding consistency. It applies uniformly to both items and users, making it especially beneficial in cold-start scenarios where either entity type may have limited interaction data.

C. Rating Distribution Consistency

To ensure fair and consistent supervision under cold-start scenarios, we design a rating consistency loss that encourages the student model to replicate the teacher's relative scoring behavior, thereby reducing discrepancies in predicted relevance

scores. Here, “rating” refers to the predicted relevance score used for top- K recommendation rather than an explicit user rating.

We match the teacher and student predictions on both positive and negative samples. For a triplet (u, i, j) , where i is a positively interacted item and j is a negative sample, the loss is defined as:

$$\begin{aligned}\mathcal{L}_{\text{rate}} &= \mathcal{L}_{\text{rate}}^I + \mathcal{L}_{\text{rate}}^J \\ &= \sum_{(u,i,j) \in \mathcal{B}} w_u w_i \left((\hat{y}_{ui}^{(s)} - \hat{y}_{ui}^{(t)})^2 + (\hat{y}_{uj}^{(s)} - \hat{y}_{uj}^{(t)})^2 \right).\end{aligned}$$

By minimizing the squared prediction differences, the student is guided to produce relevance scores that are consistent with those of the teacher, regardless of user/item cold-start status. Although alternative objectives such as KL divergence [54] or Wasserstein distance [55] can be used to compare distributions, we adopt this pointwise approach due to its simplicity and practical effectiveness.

D. Ranking Consistency

To ensure consistent ranking under cold-start scenarios, we introduce a ranking consistency loss that explicitly enforces the pairwise ranking consistency between the student and the teacher. This consistency is particularly important because the teacher utilizes rich behavioral embeddings, while the student relies solely on content information.

Given a training triplet (u, i, j) , where i is a positive item and j is a negative sample for user u , we define the ranking probabilities of the teacher and student as:

$$\begin{aligned}r_{uij}^{(t)} &= \sigma(\hat{y}_{ui}^{(t)} - \hat{y}_{uj}^{(t)}), \\ r_{uij}^{(s)} &= \sigma(\hat{y}_{ui}^{(s)} - \hat{y}_{uj}^{(s)}),\end{aligned}$$

where $\sigma(\cdot)$ denotes the sigmoid function. The ranking scores $r_{uij}^{(t)}$ and $r_{uij}^{(s)}$ represent the teacher’s and student’s probabilities of ranking item i higher than item j for user u , respectively.

To minimize the discrepancy in ranking behavior, we adopt a soft cross-entropy formulation:

$$\begin{aligned}\mathcal{L}_{\text{rank}} &= - \sum_{(u,i,j) \in \mathcal{B}} w_u w_i \left(r_{uij}^{(t)} \log r_{uij}^{(s)} \right. \\ &\quad \left. + (1 - r_{uij}^{(t)}) \log(1 - r_{uij}^{(s)}) \right).\end{aligned}$$

By minimizing this ranking loss, the student is encouraged to mimic the teacher’s pairwise ordering of items, thereby improving its ranking performance even in the absence of historical user behavior. This facilitates effective knowledge transfer from behavior-rich teachers to content-driven student models under cold-start conditions.

E. Representation Distribution Consistency

To improve the global consistency between student and teacher embeddings, we propose a distribution loss that matches their first-order (mean) and second-order (covariance) statistics. Compared to kernel-based methods such as Maximum Mean

TABLE II
EMPIRICAL EFFICIENCY ON LARGE-SCALE DATASETS

Dataset	Method	Preproc. Time (s)	Train Time (s)	Epochs	Peak GPU Mem. (GB)	Inference Time (s)
XING	LightGCN	16.81	10298.42	131	1.54	7.27
	DropoutNet	17.19	1292.79	30	1.29	7.75
	BACD	17.21	1246.84	30	0.62	7.66
Book-Crossing	LightGCN	8.43	1839.61	66	3.61	12.22
	DropoutNet	5.81	503.33	25	3.39	12.43
	BACD	6.10	559.07	35	1.54	12.44

Discrepancy (MMD), this approach is computationally efficient while effectively capturing global distribution differences.

Instead of combining users and items into a single distribution, we compute statistics separately for users, positive items, and negative items. Let $\mathcal{B}_U$, $\mathcal{B}_I$, and $\mathcal{B}_J$ represent the sampled sets of users, positive items, and negative items, respectively.

Mean and Covariance Estimation: For each $X \in \{U, I, J\}$ corresponding to users, positive items, and negative items, respectively, we compute:

$$\begin{aligned}\mu_X^{(t)} &= \frac{1}{|\mathcal{B}_X|} \sum_{x \in \mathcal{B}_X} \mathbf{e}_x, \quad \mu_X^{(s)} = \frac{1}{|\mathcal{B}_X|} \sum_{x \in \mathcal{B}_X} \tilde{\mathbf{e}}_x, \\ \Sigma_X^{(t)} &= \frac{1}{|\mathcal{B}_X|} \sum_{x \in \mathcal{B}_X} (\mathbf{e}_x - \mu_X^{(t)})(\mathbf{e}_x - \mu_X^{(t)})^\top, \\ \Sigma_X^{(s)} &= \frac{1}{|\mathcal{B}_X|} \sum_{x \in \mathcal{B}_X} (\tilde{\mathbf{e}}_x - \mu_X^{(s)})(\tilde{\mathbf{e}}_x - \mu_X^{(s)})^\top.\end{aligned}$$

The representation distribution loss is defined as:

$$\mathcal{L}_{\text{rep}} = \sum_{X \in \{U, I, J\}} \left(\|\mu_X^{(s)} - \mu_X^{(t)}\|_2^2 + \gamma \|\Sigma_X^{(s)} - \Sigma_X^{(t)}\|_F^2 \right).$$

This separation preserves the semantic distinctions among users, positive items, and negative items, yielding a more precise distribution-level alignment.

F. Complexity Analysis

We analyze BACD by separating teacher pre-training, offline preprocessing, distillation training, and inference. The one-time teacher pre-training cost has the same order as training the corresponding backbone and is reported separately from the BACD distillation cost in Table II.

Offline Preprocessing: Teacher-quality weights are computed once after teacher pre-training and are then fixed throughout distillation. Counting interaction degrees costs $O(|\mathcal{H}|)$. BACD does not materialize dense user-user or item-item similarity matrices; instead, it enumerates sparse nonzero co-occurrence candidates from adjacency lists and extracts sparse top- K_{nbr} neighbor lists. Let $|\mathcal{E}_{\text{co}}| = |\{(i, j) : i \neq j, S_{ij}^I > 0\}| + |\{(u, v) : u \neq v, S_{uv}^U > 0\}|$ denote the number of emitted co-occurrence candidates. This candidate-level construction avoids full pairwise similarity computation over all users and items. Heap-based top- K_{nbr} pruning costs $O(|\mathcal{E}_{\text{co}}| \log K_{\text{nbr}})$ and stores $O((|\mathcal{U}_w| + |\mathcal{I}_w|)K_{\text{nbr}})$ retained neighbors. Computing neighborhood deviations over the retained lists further costs $O((|\mathcal{U}_w| + |\mathcal{I}_w|)K_{\text{nbr}}d)$.

Thus, teacher-quality weighting is a one-time offline step and does not add per-epoch or inference-time overhead.

Training: For T epochs on interaction set $\mathcal{H}$, let d_c denote the maximum content-feature dimension and assume the hidden width of the two-layer content encoder is $O(d)$. With one sampled negative item per interaction, content encoding costs $O(|\mathcal{H}|d_c d)$ per epoch and teacher/student score computation costs $O(|\mathcal{H}|d)$. The rating and ranking consistency losses operate on these computed scores and add only $O(|\mathcal{H}|)$ element-wise operations per epoch. The representation distribution consistency term $\mathcal{L}_{\text{rep}}$ is computed on each stochastic training minibatch: for batch size B and embedding dimension d , covariance estimation costs $O(Bd^2)$ per batch and $O(|\mathcal{H}|d^2)$ per epoch. Since B and d are fixed hyperparameters in large-scale training, the per-batch overhead does not grow with the total number of users or items; for fixed d , the per-epoch overhead is linear in $|\mathcal{H}|$, rather than quadratic in $|\mathcal{U}|$ or $|\mathcal{I}|$. Thus, the overall distillation complexity is $O(T|\mathcal{H}|(d_c d + d^2))$, up to lower-order score and element-wise loss terms.

Inference and Memory: At inference time, BACD uses the same dot-product scoring operation as the backbone; scoring a candidate set $\mathcal{S}$ costs $O(|\mathcal{S}|d)$. Generating an uncached cold representation adds $O(d_c d)$ content-encoder cost per cold entity. Teacher-quality weighting and all distillation losses are not invoked during inference. During training, the main additional storage consists of cached teacher embeddings $O((|\mathcal{U}_w| + |\mathcal{I}_w|)d)$, retained neighbor lists $O((|\mathcal{U}_w| + |\mathcal{I}_w|)K_{\text{nbr}})$, teacher-quality weights $O(|\mathcal{U}_w| + |\mathcal{I}_w|)$, and batch-level representations/covariance statistics $O(Bd + d^2)$.

Empirical Efficiency: Table II reports preprocessing time, stage-level training time until early stopping, the corresponding number of epochs, peak GPU memory, and overall inference latency on two representative large-scale datasets. We include LightGCN as the backbone/teacher and DropoutNet as a representative cold-start baseline. All efficiency measurements are conducted on a single NVIDIA GeForce RTX 5090 GPU under the same software environment.

Table II shows that BACD remains practical in runtime and memory. The LightGCN row reports the backbone/teacher training cost, while the BACD row reports the distillation-stage cost after teacher pre-training. BACD distillation takes 1246.84 s on XING and 559.07 s on Book-Crossing, compared with 10298.42 s and 1839.61 s for LightGCN teacher training. Compared with DropoutNet, BACD has comparable training time and inference latency while requiring substantially less peak GPU memory.

V. EXPERIMENTS

We conduct comprehensive experiments on public datasets to evaluate the effectiveness of the proposed **Behavior-Aware Consistent Distillation (BACD)** framework, focusing on the following key research questions:

- **RQ1:** Can BACD outperform state-of-the-art methods in both cold-start item and cold-start user scenarios, across overall, warm, and cold-start recommendations?

- **RQ2:** How do the different components of BACD affect its performance?
- **RQ3:** Is BACD robust to noisy or partially missing content features, and does it remain effective in few-interaction cold-start adaptation beyond the strict zero-interaction setting?
- **RQ4:** Does BACD remain effective under temporal cold-start evaluation?

A. Experimental Setup

Datasets: To evaluate the effectiveness of BACD under both item and user cold-start conditions, we conduct experiments on four widely used benchmark datasets: CiteULike, XING, MovieLens, and Book-Crossing [56], [57], [58], [59].

- **Cold-Start Item Setup:** The **CiteULike** dataset contains 5,551 users, 16,980 articles, and 204,986 interactions; each article is represented by a 300-dimensional embedding from title and abstract. The **XING** dataset contains 106,881 users, 20,519 jobs, and 3,856,580 interactions; each job is represented by a 2,738-dimensional feature vector including career level, industry, and tags.
- **Cold-Start User Setup:** The **MovieLens** dataset contains 6,040 users, 3,706 movies, and 1,000,209 ratings; each user is represented by a 3,469-dimensional feature vector from gender, age, occupation, and zip code. The **Book-Crossing** dataset contains 92,107 users, 270,170 books, and 1,031,175 ratings; each user is represented by a 2,312-dimensional feature vector from age and location.

For both settings, 20% of target entities are randomly designated as cold-start. All interactions of these target cold entities are removed from the training graph and split evenly into validation and test sets, while the remaining warm interactions are divided into training, validation, and test sets with ratio 8:1:1.

Performance Metrics: We follow the full-ranking protocol in [3], [4] and report Recall@K and NDCG@K under overall, warm, and cold-start settings. Unless otherwise specified, $K = 20$.

Implementation Details: All baselines are implemented from official releases. For GAR, we use the latest official version and the CLCRec evaluation protocol in [20]. The embedding dimension is 200 for all methods. We use Adam with learning rate 0.001, batch size 1024, and weight decay 0.001, with early stopping on validation NDCG@K. Hyperparameters are selected by grid search for each dataset.

Additional Evaluation Settings: Beyond the standard zero-interaction cold-start benchmark, we add four settings for RQ3–RQ4: (i) *noisy content*, where we corrupt the content features of target cold entities at evaluation time by adding zero-mean Gaussian noise $\mathcal{N}(0, \sigma^2)$, with $\sigma \in \{0.1, 0.3, 0.5\}$ denoting the noise standard deviation; (ii) *missing content*, where we randomly mask a $\rho \in \{0.1, 0.3, 0.5\}$ proportion of feature dimensions at evaluation time; (iii) *few-interaction cold-start adaptation*, where $n_{\text{obs}} \in \{1, 3, 5\}$ observed interactions of each target cold entity are moved into the training graph and the remaining interactions are kept for validation and testing, with $n_{\text{obs}} = 0$

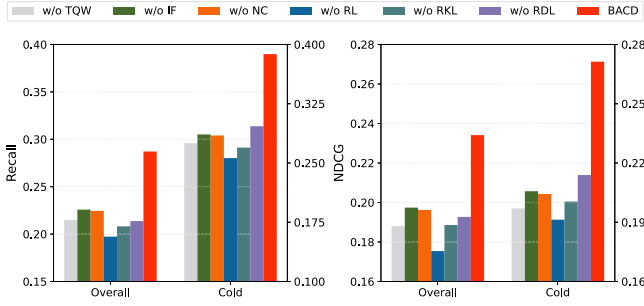


Fig. 3. Ablation study results on XING for cold-start items.

corresponding to the original zero-interaction setting; and (iv) *temporal cold-start evaluation*, where we use a chronological MovieLens split (8:1:1) and regard a user as temporally cold if the first interaction appears after the training cutoff. Unless otherwise specified, these analyses use the LightGCN backbone.

Benchmark Models: To examine the effectiveness and robustness of BACD, we benchmark it against eight representative cold-start recommendation models on the selected datasets:

- **DeepMusic** [14] applies deep neural networks to minimize the mean squared error between generated embeddings and warm embeddings.
- **MetaEmb** [12] leverages a meta-learning approach to accelerate the training convergence of generated embeddings.
- **GAR** [34] synthesizes embeddings via adversarial interaction with a warm recommendation model.
- **DropoutNet** [15] enhances cold-start resilience by randomly omitting embeddings during training.
- **MTPR** [35] creates counterfactual cold embeddings combining dropout with BPR-based ranking.
- **Heater** [11] extends DropoutNet with a mixture-of-experts design and modeling of embedding similarity.
- **CLCRec** [16] approaches cold-start recommendation using contrastive learning grounded in information-theoretic principles.
- **ALDI** [20] utilizes an aligning distillation strategy to transfer knowledge from warm to cold items, reducing the recommendation gap.

B. Key Findings (RQ1)

We compare BACD with various baselines on four datasets under overall recommendation (Eq. (1)), warm-start recommendation (Eq. (2)), and cold-start recommendation (Eqs. (3) or (4)). Tables III and IV report results for cold-start items on CiteULike/XING and cold-start users on MovieLens/Book-Crossing, respectively. We use three representative backbones: Matrix Factorization (MF) [1], Neural Graph Collaborative Filtering (NGCF) [3], and LightGCN [4].

For each backbone, we compare the base model with four generative approaches (DeepMusic, MetaEmb, GAR, ALDI) and four dropout-based methods (DropoutNet, MTPR, Heater, CLCRec). Since CLCRec and ALDI are not applicable to cold-start users, their results are excluded from Table IV. Performance gains are calculated relative to the strongest baseline for each backbone, with the best-performing values underlined.

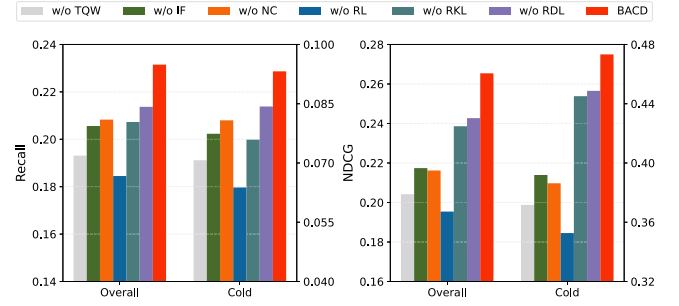


Fig. 4. Ablation study results on Book-Crossing for cold-start users.

- Across all datasets and backbones, BACD consistently ranks at the top in both overall and cold-start recommendation, benefiting from the alignment between student and teacher embeddings.
- Among individual baselines, ALDI performs best for overall recommendation in cold-start item scenarios on CiteULike and XING. In warm-start settings and cold-start user scenarios, generative methods generally outperform dropout-based alternatives.
- Models that rely on randomly initialized embeddings for items or cold-start users consistently yield the poorest results across all datasets and backbones, highlighting the critical need for effective cold-start strategies.

C. Ablation Study (RQ2)

We conduct an ablation study of our proposed BACD framework to evaluate the contributions of its key components. We evaluate the ablation variants on XING for cold-start items and Book-Crossing for cold-start users; the results are shown in Figs. 3 and 4, respectively. Specifically, we compare BACD with the following variants: (i) *w/o TQW* removes the behavior-aware teacher-quality weighting, assigning equal importance to all teacher embeddings. (ii) *w/o IF* only retains the neighborhood consistency weight, discarding the interaction-frequency weight. (iii) *w/o NC* only retains the interaction-frequency weight, discarding the neighborhood consistency weight. (iv) *w/o RL* removes the rating distribution consistent distillation objective. (v) *w/o RKL* removes the ranking consistent distillation objective. (vi) *w/o RDL* removes the representation distribution consistent distillation objective.

From the results, we can make the following observations:

Effectiveness of teacher-quality weighting: The performance decline for *w/o TQW* demonstrates the importance of evaluating teacher reliability. By adaptively emphasizing high-quality user and item embeddings, BACD can mitigate noisy supervision and improve cold-start recommendation accuracy. Comparing *w/o IF* and *w/o NC* reveals that both components contribute to teacher reliability estimation.

D. Robustness to Noisy and Partially Missing Content and Few-Interaction Adaptation (RQ3)

We evaluate BACD under noisy content, missing content, and few-interaction cold-start adaptation. For the first two settings,

TABLE III
PERFORMANCE COMPARISON UNDER THE COLD-START ITEM SCENARIO (OVERALL, COLD ITEM, AND WARM RECOMMENDATION) ACROSS THREE BACKBONE MODELS (MF, NGCF, AND LIGHTGCN). BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED IN **BOLD** AND UNDERLINED, RESPECTIVELY.

Method		Overall Recommendation				Cold Item Recommendation				Warm Recommendation			
		CiteULike		XING		CiteULike		XING		CiteULike		XING	
		Recall	NDCG	Recall	NDCG	Recall	NDCG	Recall	NDCG	Recall	NDCG	Recall	NDCG
MF	Backbone	.0776	.0647	.1444	.1276	.0056	.0031	.0037	.0021	.2838	.1933	.4505	.2946
	DropoutNet	.0794	.0670	.1733	.1454	.2268	.1356	.2773	.1953	.1343	.0792	.3304	.2140
	MTPR	.1060	.0810	.2100	.1750	.2496	.1476	.3314	.2299	.1728	.0998	.3834	.2532
	Heater	.1134	.0913	.2059	.1675	.2372	.1419	.2885	.1978	.1871	.1153	.3622	.2334
	CLCRec	.1269	.0992	.2188	.1845	.2295	.1347	.3184	.2272	.1898	.1167	.3568	.2397
	DeepMusic	.0956	.0789	.1876	.1467	.2141	.1262	.2691	.1606	.2838	.1933	.4505	.2946
	MetaEmb	.0972	.0804	.1569	.1350	.2232	.1306	.2698	.1554	.2838	.1933	.4505	.2946
	GAR	.1440	.1132	.2381	.2105	.2453	.1479	.2945	.2192	.2272	.1438	.4091	.2708
	ALDI	.1618	.1204	.2727	.2223	.2684	.1550	.3595	.2521	.2838	.1933	.4505	.2946
	BACD	.1737	.1275	.2871	.2341	.2937	.1687	.3878	.2713	.2838	.1933	.4505	.2946
%Improv.	7.35%	5.90%	5.28%	5.31%	9.43%	8.84%	7.87%	7.63%	-	-	-	-	
NGCF	Backbone	.0987	.0723	.1839	.1601	.0048	.0024	.0039	.0022	.2347	.1485	.4378	.2764
	DropoutNet	.0813	.0656	.1129	.0976	.2211	.1278	.1936	.1215	.1416	.0842	.2750	.1708
	MTPR	.1006	.0769	.1301	.1073	.2479	.1391	.1085	.0623	.1753	.0977	.3145	.1868
	Heater	.0989	.0734	.1843	.1605	.2381	.1279	.1249	.1127	.1874	.1057	.4373	.2759
	CLCRec	.1201	.0920	.1948	.1677	.2093	.1188	.2254	.1839	.1886	.1136	.4018	.2545
	DeepMusic	.1269	.1043	.2226	.1812	.1980	.1152	.2934	.2037	.2347	.1485	.4378	.2764
	MetaEmb	.1119	.0957	.1841	.1603	.2430	.1364	.2956	.2160	.2347	.1485	.4378	.2764
	GAR	.1144	.0909	.2167	.1731	.2099	.1251	.2807	.2003	.1850	.1143	.3967	.2380
	ALDI	.1541	.1141	.2333	.1912	.2466	.1394	.3007	.2235	.2347	.1485	.4378	.2764
	BACD	.1638	.1237	.2586	.2127	.2688	.1603	.3246	.2479	.2347	.1485	.4378	.2764
%Improv.	6.29%	8.41%	10.84%	11.24%	9.00%	14.99%	7.95%	10.92%	-	-	-	-	
LightGCN	Backbone	.0812	.0622	.1311	.1161	.0041	.0019	.0045	.0026	.2528	.1541	.4248	.2919
	DropoutNet	.0883	.0639	.1661	.1431	.2309	.1312	.2732	.1860	.1175	.0692	.3388	.2182
	MTPR	.1001	.0753	.1928	.1648	.2585	.1454	.3254	.1955	.1753	.0967	.3930	.2618
	Heater	.1118	.0894	.2095	.1830	.2438	.1407	.3271	.2224	.1946	.1193	.3982	.2708
	CLCRec	.1293	.0965	.2067	.1791	.2435	.1425	.3087	.2138	.2149	.1302	.3925	.2650
	DeepMusic	.0985	.0745	.1651	.1460	.2239	.1259	.2870	.1702	.2528	.1541	.4248	.2919
	MetaEmb	.0924	.0714	.1696	.1494	.2252	.1295	.2823	.1764	.2528	.1541	.4248	.2919
	GAR	.1357	.1062	.2205	.1848	.2539	.1489	.3017	.2171	.2339	.1455	.4131	.2793
	ALDI	.1626	.1201	.2409	.2041	.2692	.1539	.3377	.2356	.2528	.1541	.4248	.2919
	BACD	.1728	.1322	.2667	.2221	.2856	.1677	.3616	.2628	.2528	.1541	.4248	.2919
%Improv.	6.27%	10.07%	10.71%	8.82%	6.09%	8.97%	7.08%	11.54%	-	-	-	-	

we report LightGCN results on CiteULike and Book-Crossing against the strongest applicable baselines, namely ALDI and MetaEmb. We then evaluate few-interaction adaptation on XING for cold-start items and MovieLens for cold-start users, where n_{obs} denotes the number of observed interactions of each target cold entity moved into the training graph before evaluation. In Tables V and VI, *Baseline* reports ALDI for cold-start item datasets and MetaEmb for cold-start user datasets; best values are highlighted in **bold**.

Table V shows that BACD remains consistently more robust than the selected strongest baselines under both Gaussian perturbation and random masking. For example, at $\sigma = 0.5$, BACD achieves Recall@20 and NDCG@20 scores of 0.2727 and 0.1673 on CiteULike, compared with 0.2430 and 0.1435 for ALDI. On Book-Crossing, BACD reaches Recall@20 and NDCG@20 scores of 0.1300 and 0.0804, compared with 0.0881 and 0.0529 for MetaEmb. At $\rho = 0.5$, BACD still obtains 0.1381/0.0720 on CiteULike and 0.1282/0.0756 on Book-Crossing in terms of Recall@20 and NDCG@20, again clearly

outperforming the corresponding baselines. Table VI further shows that BACD remains effective in the few-interaction adaptation regime. On XING, BACD improves ALDI by 6.12%–7.84% in Recall and 11.88%–12.60% in NDCG when $n_{\text{obs}} \in \{1, 3\}$, and still keeps clear Recall and NDCG gains of 3.62% and 8.14% at $n_{\text{obs}} = 5$. On MovieLens, BACD consistently improves MetaEmb for cold-start users, with Recall gains of 13.46%, 11.41%, and 11.45% and NDCG gains of 8.63%, 3.55%, and 6.17% for $n_{\text{obs}} = 1, 3, 5$, respectively. These results indicate that BACD is not limited to the strict zero-interaction case; the learned behavior-aware consistency also complements sparse observed interactions in few-interaction cold-start scenarios.

E. Temporal Cold-Start Evaluation (RQ4)

Because random splits do not fully capture real deployment, we additionally evaluate BACD under a temporal cold-start protocol on MovieLens, the only dataset in our benchmark with reliable timestamps. Interactions are sorted chronologically and

TABLE IV
PERFORMANCE COMPARISON UNDER THE COLD-START USER SCENARIO (OVERALL, COLD USER, AND WARM RECOMMENDATION) ACROSS THREE BACKBONE MODELS (MF, NGCF, AND LIGHTGCN). BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED IN **BOLD** AND UNDERLINED, RESPECTIVELY.

	Method	Overall Recommendation				Cold User Recommendation				Warm Recommendation			
		MovieLens		Book-Crossing		MovieLens		Book-Crossing		MovieLens		Book-Crossing	
		Recall	NDCG	Recall	NDCG	Recall	NDCG	Recall	NDCG	Recall	NDCG	Recall	NDCG
MF	Backbone	.2044	.1999	.0701	.0344	.0053	.0409	.0019	.0008	.2293	.2199	.1121	.0554
	DropoutNet	.1560	.1931	.0271	.0123	.0675	.3919	.0168	.0092	.1670	.1679	.0299	.0126
	MTPR	.0717	.1056	.0792	.0421	.0605	.3527	.0741	.0425	.0731	.0743	.0671	.0334
	Heater	.2034	.1983	.0682	.0338	.0052	.0419	.0019	.0017	.2282	.2180	.1090	.0544
	DeepMusic	.2111	.2387	.1158	.0614	.0651	.3863	<u>.0915</u>	.0543	.2293	.2199	.1121	.0554
	MetaEmb	.2134	<u>.2483</u>	.1141	<u>.0616</u>	<u>.0825</u>	<u>.4480</u>	.0881	<u>.0550</u>	.2293	.2199	.1121	.0554
	GAR	<u>.2135</u>	.2453	<u>.1186</u>	.0570	.0798	.4451	.0830	.0371	.2289	.2193	.1116	.0529
	BACD	.2315	.2654	.1321	.0735	.0932	.4733	.1034	.0662	.2293	.2199	.1121	.0554
	%Improv.	8.43%	6.89%	11.38%	19.32%	12.97%	5.65%	13.01%	20.36%	-	-	-	-
NGCF	Backbone	.1982	.2310	.0789	.0373	.0705	.3996	.0507	.0240	<u>.2141</u>	<u>.2096</u>	<u>.0863</u>	<u>.0412</u>
	DropoutNet	.1647	.2069	.0705	.0342	.0835	.4438	.0376	.0185	.1748	.1756	.0819	.0401
	MTPR	.0796	.1126	.0792	.0421	.0598	.3604	.0741	.0425	.0820	.0813	.0671	.0334
	Heater	.1941	.2315	.0965	.0471	.0758	.4193	.0818	.0425	.2088	.2077	.0883	.0420
	DeepMusic	.1980	.2313	<u>.1050</u>	.0535	.0687	.4015	<u>.1012</u>	.0560	<u>.2141</u>	<u>.2096</u>	<u>.0863</u>	<u>.0412</u>
	MetaEmb	.1987	.2351	.1044	<u>.0538</u>	.0751	.4354	.1000	<u>.0563</u>	<u>.2141</u>	<u>.2096</u>	<u>.0863</u>	<u>.0412</u>
	GAR	<u>.2022</u>	<u>.2362</u>	.0997	.0507	<u>.0810</u>	<u>.4471</u>	.0918	.0486	.2286	.2207	.0853	.0401
	BACD	.2139	.2487	.1176	.0639	.0904	.4718	.1088	.0648	<u>.2141</u>	<u>.2096</u>	<u>.0863</u>	<u>.0412</u>
	%Improv.	5.79%	5.29%	12.00%	18.77%	8.26%	5.52%	7.51%	15.10%	-	-	-	-
LightGCN	Backbone	.1975	.2028	.0939	.0496	.0119	.0782	.0352	.0190	.2207	.2185	.1225	.0649
	DropoutNet	.1735	.2110	.0930	.0525	.0671	.3981	.0728	.0470	.1868	.1872	.0897	.0470
	MTPR	.0796	.1126	.0792	.0421	.0598	.3604	.0741	.0425	.0820	.0813	.0671	.0334
	Heater	.1958	.1936	.1085	.0553	.0002	.0017	.0632	.0310	.2202	.2180	.1223	.0645
	DeepMusic	.2041	.2407	.1236	.0713	.0705	.4159	.0929	.0622	.2207	.2185	.1225	.0649
	MetaEmb	<u>.2049</u>	<u>.2447</u>	<u>.1386</u>	<u>.0785</u>	<u>.0773</u>	<u>.4513</u>	<u>.1234</u>	<u>.0771</u>	.2207	.2185	.1225	.0649
	GAR	.2020	.2430	.1352	.0762	.0731	.4270	.1170	.0722	.2205	.2182	.1218	.0642
	BACD	.2183	.2652	.1497	.0856	.0858	.4773	.1366	.0835	.2207	.2185	.1225	.0649
	%Improv.	6.54%	8.38%	8.01%	9.04%	11.00%	5.76%	10.70%	8.30%	-	-	-	-

TABLE V
ROBUSTNESS TO NOISY AND PARTIALLY MISSING FEATURES WITH LIGHTGCN

Level	Method	CiteULike		Book-Crossing	
		Recall	NDCG	Recall	NDCG
0	Clean setting				
	Baseline	0.2692	0.1539	0.1234	0.0771
	BACD	0.2856	0.1677	0.1366	0.0835
$\sigma = 0.1$	Gaussian noise				
	Baseline	0.2650	0.1522	0.1208	0.0748
	BACD	0.2820	0.1676	0.1359	0.0832
$\sigma = 0.3$	Baseline	0.2566	0.1493	0.1053	0.0643
	BACD	0.2772	0.1667	0.1336	0.0823
$\sigma = 0.5$	Baseline	0.2430	0.1435	0.0881	0.0529
	BACD	0.2727	0.1673	0.1300	0.0804
$\rho = 0.1$	Random masking				
	Baseline	0.2299	0.1260	0.1161	0.0718
	BACD	0.2672	0.1548	0.1349	0.0819
$\rho = 0.3$	Baseline	0.1404	0.0711	0.1002	0.0612
	BACD	0.2069	0.1160	0.1332	0.0794
$\rho = 0.5$	Baseline	0.0557	0.0275	0.0833	0.0499
	BACD	0.1381	0.0720	0.1282	0.0756

split into training, validation, and test sets with ratio 8:1:1; a user is temporally cold if the first interaction appears after the

TABLE VI
FEW-INTERACTION COLD-START ADAPTATION WITH LIGHTGCN

n_{obs}	Method	XING		MovieLens	
		Recall	NDCG	Recall	NDCG
0	Baseline	0.3377	0.2356	0.0773	0.4513
	BACD	0.3616	0.2628	0.0858	0.4773
	%Improv.	7.08%	11.54%	11.00%	5.76%
1	Baseline	0.3564	0.2431	0.1032	0.4751
	BACD	0.3783	0.2737	0.1171	0.5161
	%Improv.	6.12%	12.60%	13.46%	8.63%
3	Baseline	0.3695	0.2486	0.1314	0.4910
	BACD	0.3984	0.2781	0.1463	0.5084
	%Improv.	7.84%	11.88%	11.41%	3.55%
5	Baseline	0.3919	0.2593	0.1474	0.4941
	BACD	0.4061	0.2804	0.1642	0.5246
	%Improv.	3.62%	8.14%	11.45%	6.17%

training cutoff. In Table VII, best and second-best values are highlighted in **bold** and underlined, respectively.

Table VII shows that BACD remains effective under the chronological evaluation protocol, improving the best non-BACD result in each metric by 4.98% and 6.83% in overall Recall@20 and NDCG@20, and by 14.21% and 7.25% under cold-start recommendation.

TABLE VII
TEMPORAL COLD-START USER PERFORMANCE ON MOVIELENS WITH
LIGHTGCN

Method	Overall Rec.		Cold Rec.	
	Recall	NDCG	Recall	NDCG
Backbone	0.2150	0.2117	0.0063	0.0423
DropoutNet	0.1739	0.2059	0.0699	0.3687
MTPR	0.0813	0.1113	0.0611	0.3304
DeepMusic	0.2122	0.2385	0.0700	0.3696
MetaEmb	0.2231	0.2457	0.0739	0.4137
GAR	0.2224	0.2488	0.0703	0.3725
BACD	0.2342	0.2658	0.0844	0.4437
%Improv.	4.98%	6.83%	14.21%	7.25%

Summary of RQ3–RQ4: Overall, BACD remains robust under noisy or partially missing content features, effective in few-interaction cold-start adaptation, and competitive under temporal cold-start evaluation.

VI. CONCLUSION

In this paper, we propose the Behavior-Aware Consistent Distillation (BACD) framework to address the cold-start challenge in recommendation systems. BACD bridges the gap between cold-start and warm entities by transferring behavioral knowledge from behavior-rich items to cold-start items and users. It employs a behavior-aware teacher-quality weighting mechanism to prioritize reliable supervision, which enhances the precision of knowledge transfer. Additionally, three consistency-driven distillation objectives are introduced: rating distribution consistency, ranking consistency, and representation distribution consistency. These objectives help reduce discrepancies between warm and cold-start recommendations.

Extensive experiments on multiple benchmark datasets demonstrate that BACD consistently outperforms existing cold-start recommendation methods in both cold-start item and user scenarios. Ablation studies further confirm the contribution of each distillation objective to performance improvements. BACD’s unified framework offers an effective solution for handling both cold-start items and users, showing robustness and effectiveness in real-world cold-start recommendation tasks.

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